GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY
FACULTY OF COMPUTING

INTAKE 41/42



Essentials of Artificial Intelligence

(IT3182)

PROJECT PROPOSAL

AI-BASED INTELLIGENT HEALTH STATUS CLASSIFICATION AND EARLY RISK PREDICTION SYSTEM USING VITAL SIGNS

GROUP 04

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Plagiarism Declaration

We recognize that plagiarism is defined as taking someone else's ideas, work or information without giving it proper credit. We are aware of the significance of proper referencing, citing and ethical academic writing. We confirm we have followed the referencing guidelines and that our work will not be copied or presented by others as theirs.

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1. Introduction

Healthcare monitoring is important to identify any change in a person's physical condition by monitoring vital signs such as heart rate, blood oxygen saturation (SpO_2) and body temperature. However, manual monitoring and simple threshold based systems may not be effective in identifying complex trends or gradual changes in health conditions. Therefore, this project proposes an AI-Based Intelligent Health Status Classification and Early Risk Prediction System Using Vital Signs that uses Artificial Intelligence and Machine Learning to analyse vital-sign data, classify the current health status, and identify early signs of increasing health risk. The system will also deliver easily understandable risk levels, health trends and alerts to help healthcare professionals and users better track health conditions.

2. Background of the Selected Problem

Physiological state information from a person can be obtained from vital signs (heart rate, blood oxygen saturation (SpO2), body temperature). Conventional health monitoring relies on manual observation or simple threshold-based systems, which demand a lot of human resources and may not be able to identify the connections between health indicators or a gradual decline in health. One abnormal measurement may not clearly indicate a health risk, but a combination of increasing heart rate, decreasing SpO2 and increasing body temperature may provide more meaningful information. Hence, the project proposes an AI-Based Intelligent Health Status Classification and Early Risk Prediction System Using Vital Signs which uses machine learning, temporal analysis, and intelligent risk reasoning to analyze the current and historical vital-sign data, classify the health conditions and identify early signs of increasing health risk.

3. Problem Statement

Current techniques for monitoring vital signs often rely on manual observation and simple threshold-based alarms. These methods can identify obvious abnormal readings but may struggle with complex relationships between multiple vital signs or slow deterioration over time.

Hence, there is a need to develop an AI-based system for analyzing vital-sign data, classifying the health status at present and predicting an increase in health risk at an early stage. The proposed system aims to provide an intelligent decision support by combining multiple

vital signs with historical trends to generate a health-status classification and an early-risk prediction

4. Aim of the Project

To design and develop an AI-based intelligent system for analyzing vital-sign data to classify a person's current health status and predict early health risks using machine learning and intelligent decision-making techniques.

5. Specific Objectives

The proposed system will achieve the following objectives:

- **Vital-Sign Data Collection**
  To collect and store heart rate, SpO_2, body temperature, timestamps, and historical measurements.
- **Health Status Classification**
  To create a supervised machine-learning model to classify the current health status as Normal, Warning or Critical.
- **Early-Risk Prediction and Trend Detection**
  To analyze historical vital-sign data, identify changes and abnormal trends, and predict whether health risk Stable, Increasing, or Decreasing.
- **Intelligent Risk Assessment**
  To use fuzzy logic to combine multiple physiological indicators and calculate an overall health-risk level.
- **Explainable Output and Dashboard**
  To provide understandable AI results through a web-based dashboard displaying vital signs, health classification, risk prediction, historical trends, and alerts.
- **AI Model Evaluation**
  To assess the accuracy, precision, recall, f1 score and confusion matrix of the AI models
- **Ethical Considerations**
  To consider patient privacy, bias, reliability, accuracy, human responsibility, and misuse of AI during system development.

6. Target Users

The proposed system is intended to support:

- **Healthcare Workers**
  Healthcare workers can use the dashboard to monitor multiple individuals and identify those requiring attention.
- **Nurses and Caregivers**
  The system can provide warnings when abnormal patterns are identified.
- **Patients**
  Patients may use a simplified interface to view their measurements and general risk status.
- **Healthcare Organizations**
  Small healthcare centers can use the system as a prototype decision-support platform.
- **Researchers and Students**
  The system can serve as a demonstration platform for applying AI techniques to healthcare data.

7. Proposed AI Techniques

7.1 Supervised Machine-Learning Classification

The current health state will be classified as Normal, Warning or Critical by a supervised machine-learning model using Heart rate, SpO2 and body temperature as input. Various algorithms like Logistic Regression, Decision Tree, Random Forest and SVM can be compared and the best algorithm will be chosen based on the performance.

7.2 Temporal / Trend Analysis

Temporal and trend analysis will analyze the historical vital-sign measurements to identify changes and gradual health deterioration. Features like rate of change, moving averages and trends will determine if the health-risk trend is Stable, Increasing or Decreasing. If there is enough sequential data and development time, an LSTM model could be researched as an alternative approach to analyze longer term patterns.

7.3 Fuzzy Logic

A fusion of multiple vital signs and their trends will be done using fuzzy logic for an overall health-risk assessment. It can not only have a normal or abnormal value but it can also be risk assessed as Low, Moderate, High or Very High.

8. Justification for Selecting the AI Techniques

The current health status of the person is classified by learning the relationships between multiple vital signs using Supervised Machine Learning.

Temporal / Trend Analysis is selected for analyzing data from the past to detect slow trends or decline in vital signs not visible from a single measurement Fuzzy Logic is chosen to handle the uncertainty and combine multiple health information to provide an understandable total risk

9. Expected System Inputs and Outputs

Inputs	**Processing**	**Outputs**
Heart rate, SpO_2, temperature	ML Classification	Normal, Warning, Critical
Historical vital-sign data	Temporal / Trend Analysis	Stable, Increasing, Decreasing risk trend
Historical trends of vital-signs	Risk Prediction	Early-risk prediction, Risk probability
Multiple vital signs	Fuzzy Logic	Low, Moderate, High, Very High risk
Abnormal or increasing-risk patterns	Alert Logic	Warning, Critical alerts

10. Proposed tools and technologies

Frontend

- React.js
- HTML & CSS

- JavaScript
- Bootstrap

Backend

- Python
- Flask / FastAPI
- REST API
- Flask-CORS

AI / Machine Learning

- Python
- Pandas & NumPy
- Scikit-learn
- Matplotlib & seaborn
- Scipy, h5py & Joblib
- TensorFlow/Keras if LSTM is implemented

Database

- PostgreSQL
- MySQL
- Firebase

Development Tools

- Visual Studio Code
- Git & GitHub
- Jupyter Notebook
- Postman

Optional Hardware

- ESP32 and MAX30102
- Temperature sensor

11. Limitations

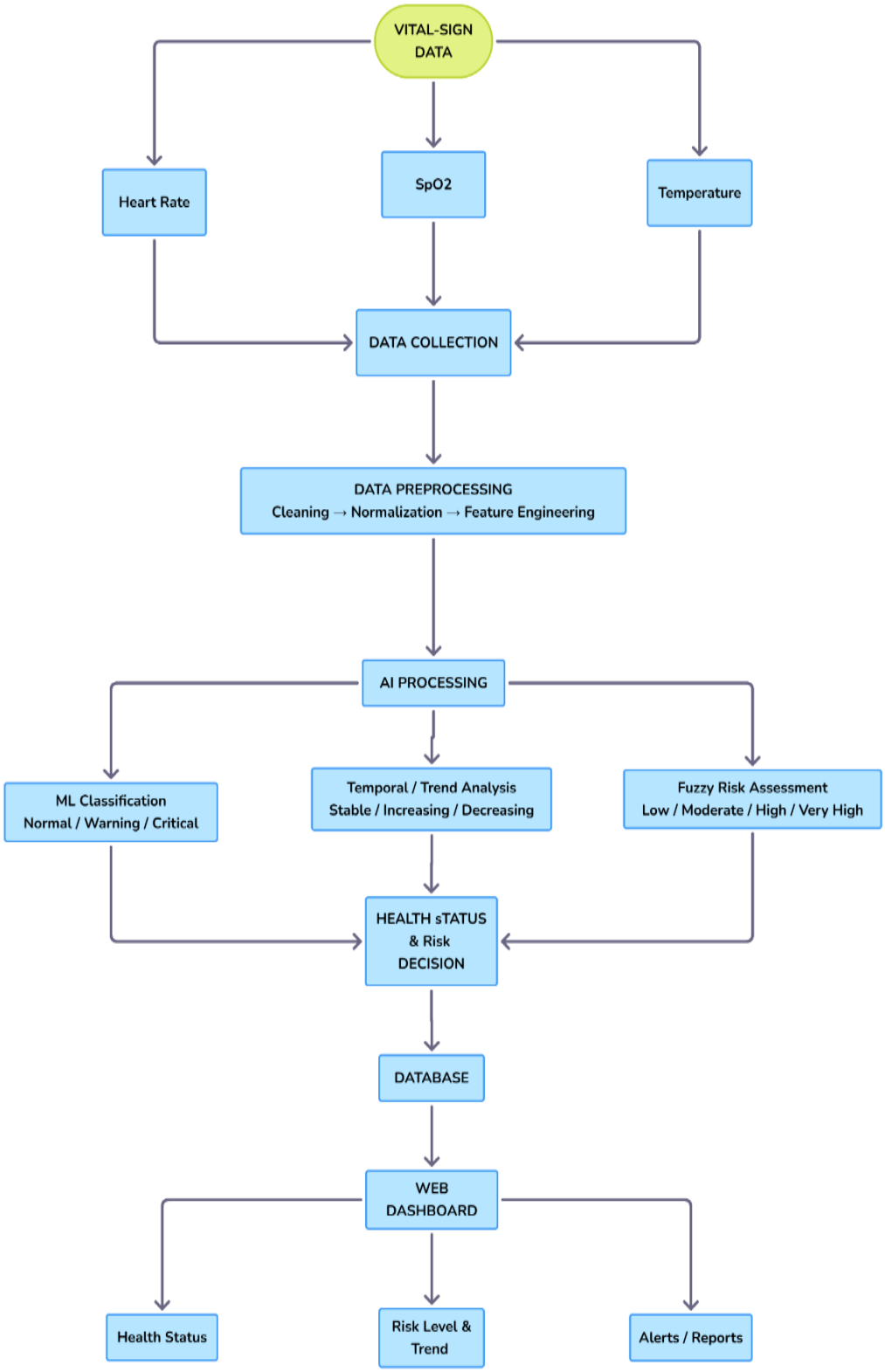
- Not intended for medical diagnosis or to replace a health care professional.
- Data quality, dataset size, synthetic data and sensor noise can impact accuracy.
- There can be false positive and false-negative predictions.
- The prototype will not be clinically certified.
- The project does not include implementing EHR or large-scale deployment.

12. Work Division Among Members

All members will take part in the problem analysis, literature review, proposal development, system design, the testing and evaluation of the AI model, system integration, testing and progress review, final documentation, and presentation.

Member	Main Area	Responsibilities
R. Dineth Meditha	ML Classification & AI Model Development	Dataset preparation, preprocessing, feature selection, ML model development, Training, Testing, Performance evaluation and Model improvement.
K.H. Prabashwari Kavindi	AI Prediction Integration, Frontend & Dashboard	APIs for AI predictions, testing, validating, displaying health classification and risk predictions, React, graphs, alerts and UI testing.
H.K.K. Aberathna	Backend & Database	Develop REST API, manage databases, patient/ vital-sign records, authentication & system integration.
T.S. Karunrathne	Risk Prediction & Fuzzy Logic	Time-series analysis, development of trend features, early-risk prediction, fuzzy variables and rules, and overall risk assessment.

13. Initial System Diagram / Workflow



14. Time Plan

Activity		Year									
		W1	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**	**W9**	**W10**
Planning & Research	Project planning										
	Problem analysis										
	Literature review										
	Proposal development										
Data & AI Model Development	Dataset selection										
	Data preprocessing										
	Classification model										
	Risk prediction										
	Fuzzy logic										
System Design & Development	Backend development										
	Frontend development										
	AI integration										
Testing & Evaluation	System testing										
	AI evaluation										
Finalization & Presentation	Final report										
	Presentation										

15. Conclusion

The Proposed AI-Based Intelligent Health Status Classification and Early Risk Prediction System Using Vital Signs utilizes Artificial Intelligence for improved health monitoring. The system will analyze the heart rate, SpO2, body temperature and previous vital sign data. Supervised Machine Learning will classify the health condition into Normal, Warning and Critical. Time-series evaluation will identify the increasing health risks, and fuzzy logic will generate an overall risk assessment. The system can provide more intelligent decision support by combining these AI techniques rather than simply monitoring with threshold values. The final deliverables will include a working prototype, AI models, sample dataset, testing results, web-based dashboard and documentation. The system is an academic prototype, and is meant to support, not replace, professional healthcare judgment.

16. References

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